

Assignment 2

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GPU: a30

b) baseline code without memory coalescing runtime: 489 ms

Code with memory coalescing runtime: 444 ms

Speedup: $489/444 = 1.101$

c) baseline code with $T = 200$

runtime = 1494089.877595 ms

parallelized code with $T = 200$

runtime : 36427.144747 ms

speedup: 41x times

With tile size $T = 200$, the achieved speedup using wavefront parallelism was 41x. This speedup is significant due to the fact that we process multiple sequence pairs in parallel (every pair has a dedicated block assigned to it) and further exploit parallelism within the block by assigning multiple threads for each diagonal within a tile so that each cell has a thread assigned to it. This also contributes to the increase in speedup. In addition, shared memory has been used instead of global memory which has a lower access time and further contributes towards the speedup.

d) The max tile size we can set is 215. Tile sizes > 215 lead to shared memory overflow errors because of limits on how much shared memory is allowed per block.

The tiling algorithm is more suitable for GPUs and FPGAs because it restricts computation to small $T \times T$ tiles that fit in fast on-chip memory, significantly reducing global memory usage.. Within each tile, wavefront parallelism enables efficient parallel computation that maps to different GPU thread blocks and FPGA pipelines. Unlike full-matrix alignment, tiling avoids storing the entire DP matrix, and unlike banded alignment, it does not impose a fixed diagonal constraint, making it both hardware-efficient and flexible.

e) Tile Size (T) = 215

Overlap (O) = 3

Runtime: 41582 ms

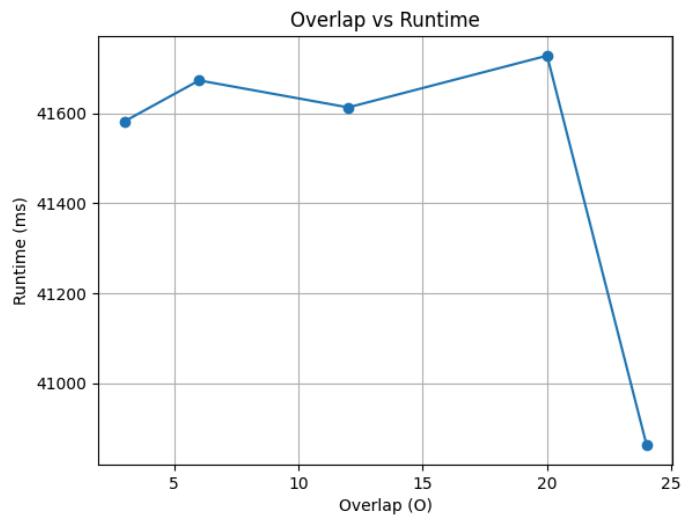
Score loss: 7.27887%

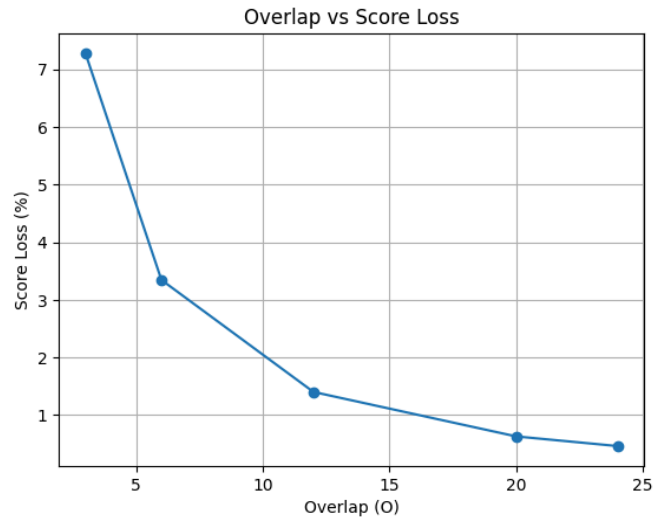
Overlap (O) = 6
Runtime: 41673 ms
Score Loss: 3.34367%

Overlap (O) = 12
Runtime: 41613 ms
Score Loss: 1.4%

Overlap (O) = 20
Runtime: 41728 ms
Score Loss: 0.628522%

Overlap (O) = 24
Runtime: 40863 ms
Score Loss: 0.460875%





It is observed that a larger overlap results in better accuracy as it increases the chances of the optimal traceback path occurring within that overlap region. This allows for more accurate estimation of traceback paths. The runtime also increases with an increase in overlap size as every tile now has to recompute the overlap region already computed by the previous tile.

f)

Overlap = 3

T = 20

Runtime: 22492 ms

Score Loss: 27.7615%

T = 40

Runtime: 22693 ms

Score Loss: 22.3468%

T = 100

Runtime: 28806 ms

Score Loss: 12.3414%

T = 150

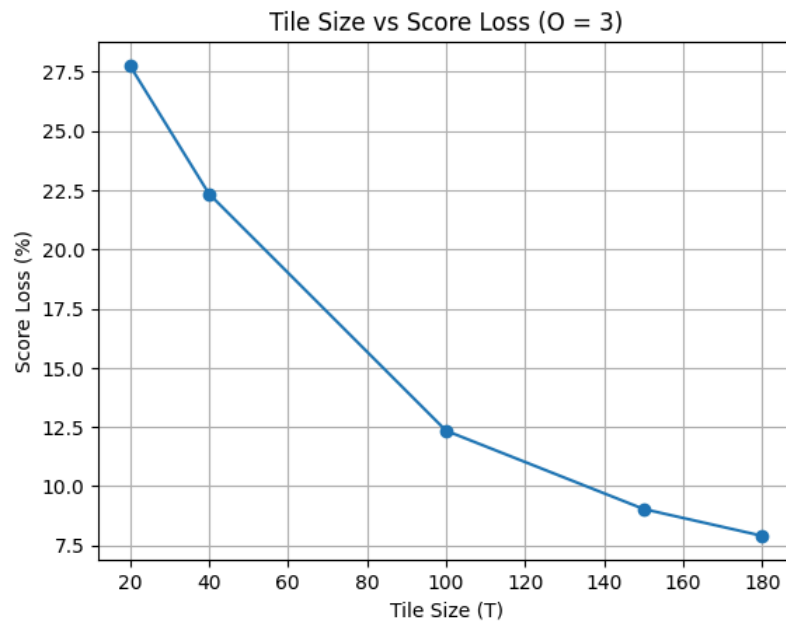
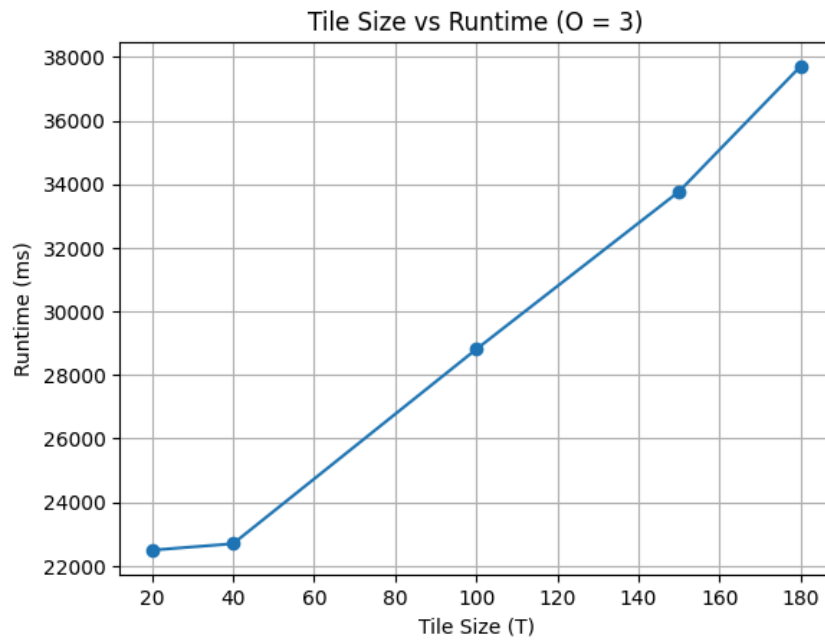
Runtime: 33762 ms

Score Loss: 9.0339%

T = 180

Runtime: 37701 ms

Score Loss: 7.90137%



It is observed that larger tile sizes result in increased accuracy because they allow more scope for capturing the optimal alignment path even if it deviates from the main diagonal. A larger tile size

increases the runtime as each tile requires computing a larger $T \times T$ DP matrix, and the number of computed cells grows quadratically with T .

g) $T = 215$, $O = 50$, numBlocks = 1024, blockSize = 32, batchsize = 2, numThreads = 8

Total Runtime: 40849 ms

Score Loss: 0.167556%

```
Setting CPU threads to 4 and printing GPU device properties.
Device Number: 0
  Device name: NVIDIA A30
  Device memory: 25231032320
  Memory Clock Rate (KHz): 1215000
  Memory Bus Width (bits): 3072
  Peak Memory Bandwidth (GB/s): 933.120000

Program completed in 40849 ms

[SUCCESS] All 10000 aligned sequences match their raw originals.

Valid Comparisons: 5000 pairs. Average Percentage Score Loss: 0.167556%
```

h) Compute Sanitizer:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

  Memory Clock Rate (KHz): 1215000
  Memory Bus Width (bits): 1536
  Peak Memory Bandwidth (GB/s): 466.560000

Program completed in 952383 ms

===== ERROR SUMMARY: 0 errors
[SUCCESS] All 10000 aligned sequences match their raw originals.

Valid Comparisons: 5000 pairs. Average Percentage Score Loss: 0.167556%

pod "a4kapoor-3549245" deleted
a4kapoor@codeserver:~/assgn2-akan1404$
```

Runtime with Compute Sanitizer: 952383 ms

Runtime without Compute Sanitizer: 40849 ms

Speedup: 22.8

i)

Since the maximum tile size we can set on our GPU is limited to 215, we may not always capture the most optimal subsequences for very long sequences, especially if the overlap O is small.

However, for sequences with high identity (98%), the optimal path stays close to the diagonal, so even smaller tiles can still achieve near-optimal accuracy. The wavefront parallelization implemented on the GPU explores all cells within a tile and its overlap, offering more flexibility in coverage than a standard banded alignment. With sufficiently large T and O , the GACT implementation can sometimes achieve a higher alignment score than the banded reference.

Ideally, the tile size should be at least as large as the expected maximum drift from the diagonal.

For example, for a sequence of length 1000 with 98% identity, a tile size of ~ 20 could cover the possible drift around the diagonal, and the overlap should be large enough to ensure that the next tile has accurate information in case the optimal path deviates from the diagonal.